

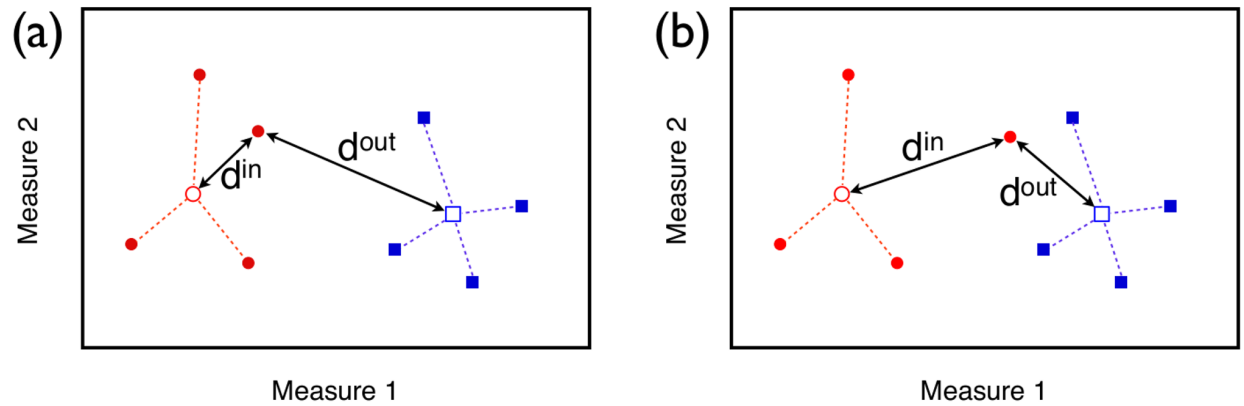
Supporting Information Text S1

Outlier analysis of cell-line replicates: centroid distances in multidimensional miR-measurement space

MiR expression was measured on 7 AML cell lines (HL60, K562, KASUMI1, KG1a, ML2, MO7e1, and U937), 9 B-ALL cell lines (KASUMI2, KOPN8, MHHCALL3, MHHCALL4, MUTZ5, NALM6, RCHACV, REH, and SUPB15), and 5 T-ALL cell lines (CCRF_CEM, JURKAT, KARPAS45, MOLT16, and MOLT3). Each cell line has been measured several times (3 to 4 replicates per cell line). In order to ensure that our machine-learning analysis is based on robust data, we developed the following robustness test to identify possible outliers.

Let us assume that, for a given disease class, n_s samples have been measured from n_{cl} different cell lines (e.g. for AML, $n_s=24$ samples have been measured from $n_{cl}=7$ different cell lines). Measured samples are labeled by subindex s ($s=1, \dots, n_s$), while different cell lines are labeled by subindex α ($\alpha=1, \dots, n_{cl}$). Since samples are described by sets of N miR expression values, they can be represented as dots in N -dimensional space (in our case, $N=370$). We renormalize the distribution of measured values for each miR (across the n_s samples) to have zero mean and unit variance. For each sample, we measure: 1) the Euclidean distance (in 370-D miR measurement space after renormalization) between that sample and the centroid of the remaining replicates for the same cell line, d_s^{in} ; 2) the Euclidean distance between that sample and the centroid of each of the other cell lines, $d_{s,\alpha}^{out}$. Notice that the centroid of a group of objects in N -dimensional space is the mean position obtained by averaging the coordinate vectors over all the objects in the group, akin to the definition of the center of mass in physics, if all the objects are of equal mass. Supporting Information Figure S1 shows a schematic representation of the definition of the distances d_s^{in} and $d_{s,\alpha}^{out}$. The distance ratio (in logarithmic scale) between replicate r and cell line α is then $R_{s,\alpha} = \log(d_{s,\alpha}^{out}/d_s^{in})$. Supporting Information Figures S2-S4 show distance ratios $R_{r,\alpha}/(R_{r,\alpha})_{max}$, which are normalized to have a maximum value of 1.

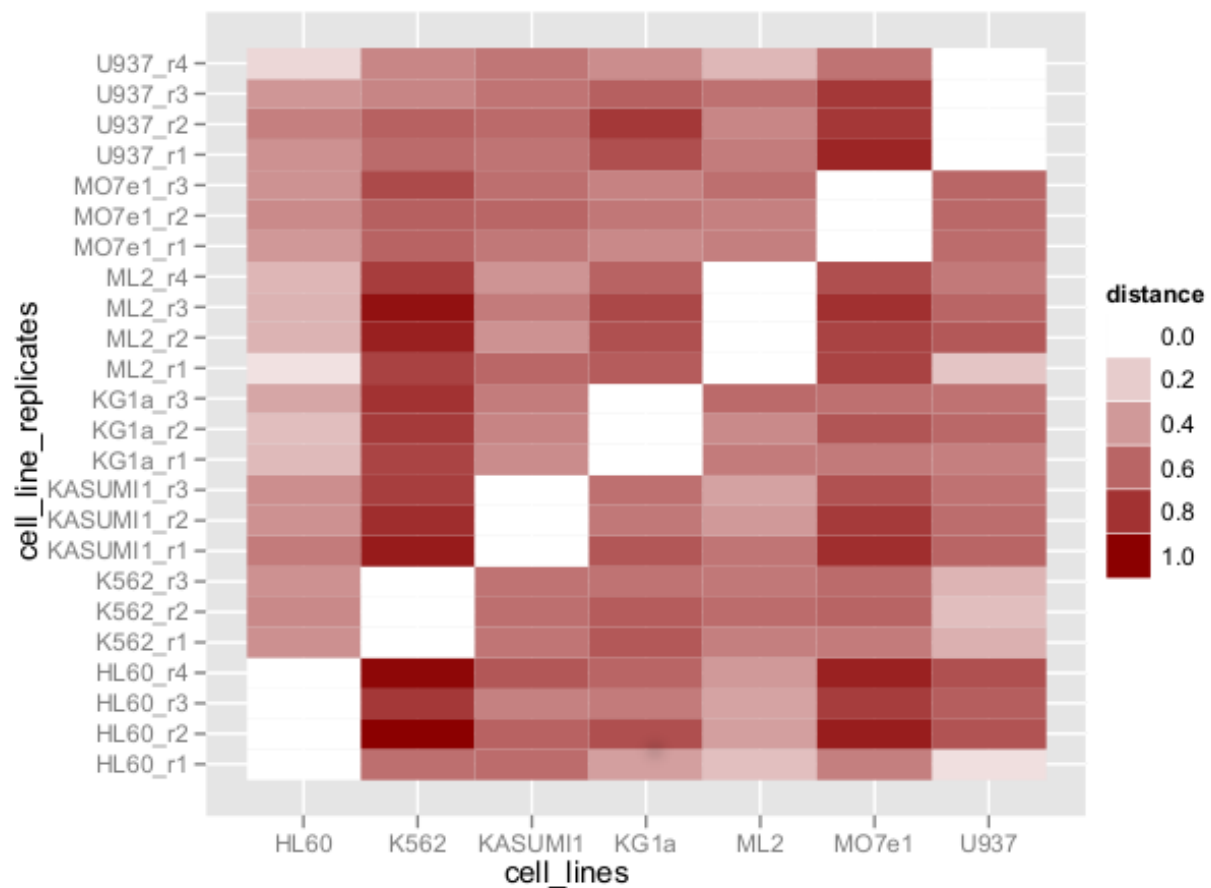
Supporting Information Figure S2 shows outlier analysis results for AML cell line data, where all normalized distance ratios between 24 cell line samples (vertical axis) and 7 different cell lines (horizontal axis) were calculated based on Euclidean distances in the 370-dimensional miR measurement space. The block-diagonal structure indicates that each sample lies closer to the centroid of the remaining replicates from the same cell line, relative to its distance to the centroid of each of the other cell lines. Therefore, we conclude that all AML replicates pass the robustness test. Supporting Information Figure S3 shows results for B-ALL cell line data (36 samples from 9 different cell lines), while Supporting Information Figure S4 displays results for T-ALL (20 samples from 5 different cell lines). In all cases, we observe similar block-diagonal matrix values that indicate that none of the measured samples should be dismissed as outlier.



CAPTION TO SUPPORTING INFORMATION FIGURE S1:

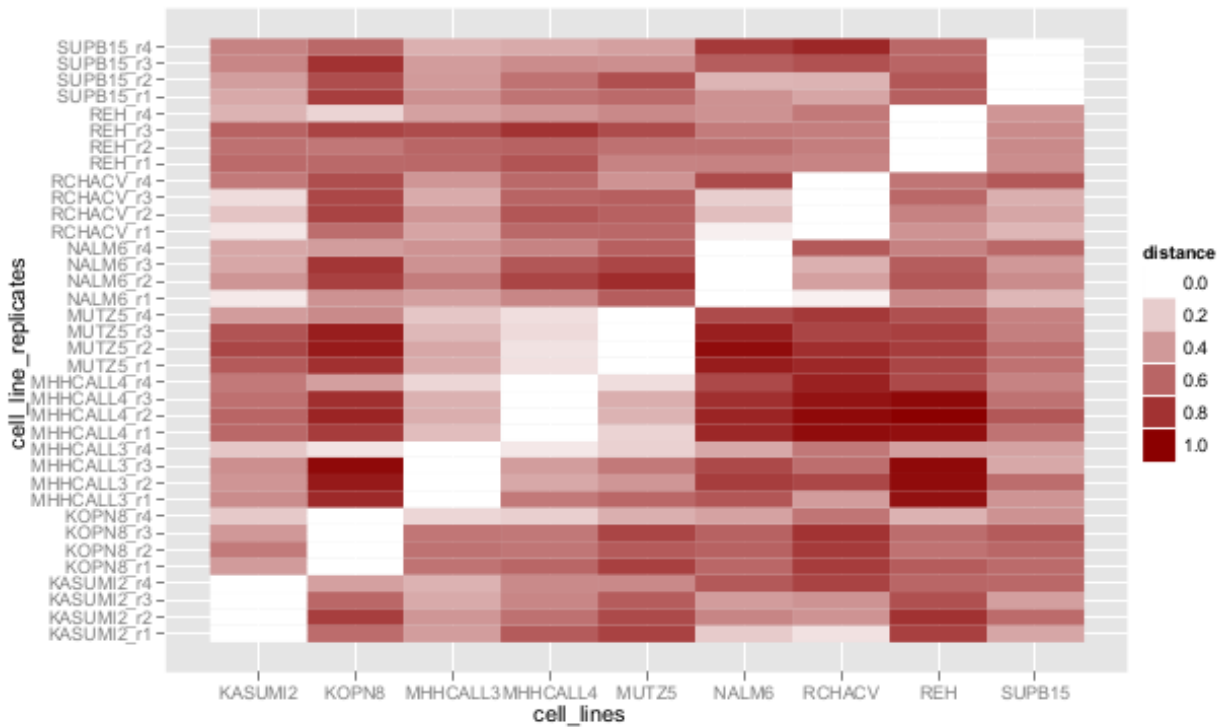
Schematic representation of the centroid-based method for outlier analysis.

Assuming for simplicity a 2-dimensional measurement space, each filled symbol represents a sample. Red circles represent replicates from cell line 1, blue squares represent replicates from cell line 2. The centroids for each group are shown by open symbols. The distance between the test sample and the centroid obtained from the remaining replicates of cell line 1 is labeled d^{in} . The distance between the test sample and the centroid obtained from all replicates of cell line 2 is labeled d^{out} . In **(a)**, $d^{\text{in}} < d^{\text{out}}$ and the chosen sample passes the robustness test. In **(b)**, $d^{\text{in}} > d^{\text{out}}$ and the chosen sample fails the test, therefore it is considered an outlier in the dataset.



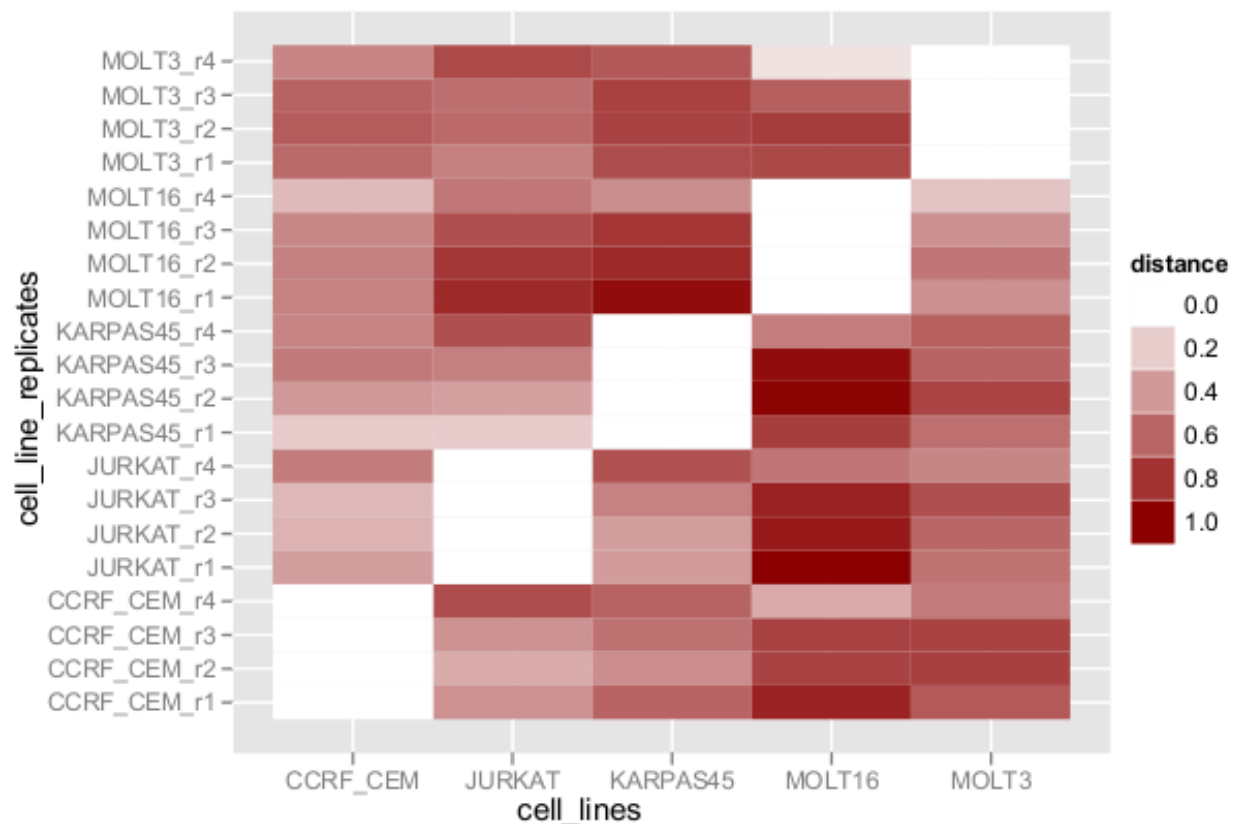
CAPTION TO SUPPORTING INFORMATION FIGURE S2:

Outlier analysis of AML cell line data: 24 replicates from 7 cell lines. Normalized distance ratios between cell line replicates (vertical axis) and cell lines (horizontal axis) based on Euclidean distances in the 370-dimensional miR measurement space. The block-diagonal structure indicates that each replicate lies closer to the center of mass of the remaining replicates from the same cell line, relative to its distance to the center of mass of each of the other cell lines.



CAPTION TO SUPPORTING INFORMATION FIGURE S3:

Outlier analysis of B-ALL cell line data: 36 replicates from 9 cell lines. Normalized distance ratios between cell line replicates (vertical axis) and cell lines(horizontal axis) based on Euclidean distances in the 370-dimensional miR measurement space. As in Fig. S2, the block-diagonal structure indicates that all replicates pass the robustness test.



CAPTION TO SUPPORTING INFORMATION FIGURE S4:

Outlier analysis of T-ALL cell line data: 20 replicates from 5 cell lines. Normalized distance ratios between cell line replicates (vertical axis) and cell lines (horizontal axis) based on Euclidean distances in the 370-dimensional miR measurement space. As in Figs. S2 and S3, the block-diagonal structure indicates that all replicates pass the robustness test.